

Supplementary Material for MedicFC

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1 Evidence Coverage Analysis

This section provides a detailed analysis of the evidence coverage achieved by the evidence retrieval and hybrid reranking pipeline. We evaluate the ability of the retrieved evidence to recover the gold evidence associated with each claim.

We report coverage at different retrieval depths and semantic similarity thresholds. Specifically, we consider the top-3, top-5, and top-10 retrieved evidence items using similarity thresholds of 0.50, 0.60, and 0.80.

For each configuration, we report the average, median, and standard deviation of evidence coverage, together with the number of claims that are fully covered, partially covered, or have zero coverage.

1.1 Coverage of Extracted Evidence

Table 1 reports the coverage of gold evidence by the evidence extracted through our retrieval pipeline after hybrid reranking.

The highest average coverage is obtained with the top-10 configuration at a similarity threshold of 0.50, achieving 73.16% average coverage and a median coverage of 100.00%. This configuration fully covers 381 claims.

1.2 Coverage of Translated Evidence

We additionally evaluate translated evidence extracted from the URLs provided with HealthFC. Table 2 reports the gold evidence coverage obtained after hybrid reranking.

For the translated evidence, the highest average coverage is obtained with top-10 evidence at a similarity threshold of 0.50, achieving an average coverage of 62.63%. This configuration fully covers 360 claims.

2 Ablation Study

We conduct an ablation study to quantify the contribution of the background information and the confidence-based evidence grouping components in MedicFC. We evaluate four configurations: the complete MedicFC pipeline, a variant without background information, a variant without confidence-based evidence grouping, and a variant without both components. The experiments are conducted using DeBERTa-v3-Large and PubMedBERT, with results reported for the Supported, Refuted, and NEI classes as well as the macro average.

Table 1. Gold evidence coverage of our extracted evidence after hybrid reranking for different retrieval depths and similarity thresholds.

Top-K	Threshold	Avg. (%)	Median (%)	Std. (%)	Full	Partial	Zero
3	0.50	65.69	75.00	–	316	307	127
3	0.60	49.05	50.00	–	178	362	210
3	0.80	9.59	0.00	–	18	150	582
5	0.50	69.12	80.00	–	338	308	104
5	0.60	52.61	50.00	–	196	372	182
5	0.80	11.39	0.00	–	21	173	556
10	0.50	73.16	100.00	34.18	381	281	88
10	0.60	58.06	66.67	–	239	354	157
10	0.80	14.30	0.00	–	26	208	516

Table 2. Gold evidence coverage of translated evidence extracted from the HealthFC-provided URLs after hybrid reranking for different retrieval depths and similarity thresholds.

Top-K	Threshold	Avg. (%)	Median (%)	Std. (%)	Full	Partial	Zero
3	0.50	57.90	66.67	–	302	227	221
3	0.60	46.62	50.00	–	190	300	260
3	0.80	16.43	0.00	–	21	261	468
5	0.50	60.13	75.00	–	331	206	213
5	0.60	50.53	55.00	–	222	287	241
5	0.80	22.42	0.00	–	35	319	396
10	0.50	62.63	80.00	–	360	183	207
10	0.60	55.36	66.67	–	268	258	224
10	0.80	31.56	25.00	–	67	368	315

Table 3 presents the class-wise results. The complete pipeline achieves a macro-F1 of 0.477 with DeBERTa and 0.499 with PubMedBERT. For PubMedBERT, removing background information reduces the macro-F1 from 0.499 to 0.473, indicating that the biomedical background information provides useful complementary evidence for verification. Removing evidence grouping results in a macro-F1 of 0.499, which is comparable to the full configuration, although the class-wise performance changes substantially. In particular, the Supported-class F1 decreases from 0.527 to 0.486, while the Refuted-class F1 increases from 0.709 to 0.743.

For DeBERTa, removing background information decreases the macro-F1 from 0.477 to 0.429. Removing grouping produces a macro-F1 of 0.472, which is close to the full configuration, but again leads to considerable changes in class-wise performance. The Refuted-class F1 increases from 0.707 to 0.759, whereas

Table 3. Ablation study of MedicFC with class-wise and overall performance.

Configuration	Class	DeBERTa			PubMedBERT		
		Precision	Recall	F1	Precision	Recall	F1
Full pipeline	Supported	0.483	0.510	0.496	0.476	0.589	0.527
	Refuted	0.715	0.700	0.707	0.742	0.678	0.709
	NEI	0.230	0.224	0.227	0.274	0.248	0.261
	Macro Avg.	0.476	0.478	0.477	0.497	0.505	0.499
w/o bg Info	Supported	0.443	0.480	0.461	0.483	0.550	0.514
	Refuted	0.672	0.759	0.713	0.728	0.638	0.680
	NEI	0.189	0.080	0.112	0.208	0.248	0.226
	Macro Avg.	0.434	0.440	0.429	0.473	0.479	0.473
w/o grouping	Supported	0.566	0.555	0.560	0.566	0.426	0.486
	Refuted	0.682	0.856	0.759	0.724	0.764	0.743
	NEI	0.333	0.056	0.096	0.243	0.296	0.267
	Macro Avg.	0.527	0.489	0.472	0.511	0.495	0.499
w/o bg Info + grp.	Supported	0.475	0.619	0.538	0.493	0.510	0.501
	Refuted	0.702	0.801	0.748	0.751	0.650	0.697
	NEI	0.250	0.008	0.016	0.240	0.336	0.280
	Macro Avg.	0.476	0.476	0.434	0.495	0.499	0.493

the NEI-class F1 decreases from 0.227 to 0.096. When both background information and grouping are removed, the macro-F1 decreases further to 0.434 for DeBERTa and 0.493 for PubMedBERT.

Overall, the ablation results indicate that background information provides a consistent contribution to verification performance, particularly for DeBERTa. The effect of confidence-based grouping is more nuanced: its removal does not necessarily reduce the macro-F1 substantially, but it changes the distribution of performance across the three classes. This is particularly evident for the NEI class, where DeBERTa’s F1 decreases from 0.227 in the full configuration to 0.096 without grouping. These results suggest that the different components of MedicFC influence not only aggregate performance but also the model’s ability to distinguish between individual verification classes.